Product Requirements Document (PRD)

E-Munshi — AI-Powered Financial Intelligence Platform for Tally ERP

Document Reference: PRD-EMUNSHI-2026-V1.0
Target Delivery Window: 30 Calendar Days from Project Kickoff
Classification: Confidential · Client & Engineering Review
Last Updated: October 26, 2026

Executive Document Control

Metadata Field	Value / Description
Product Name	**E-Munshi** (AI Financial Intelligence for Tally ERP)
Document Version	1.0 (Production Pilot Release Candidate)
Engineering Lead	Mr. Aaquib Nawab
Client Reviewers	Executive Product Sponsors, Practicing CA Advisory Board
Core Architecture	Next.js 15 (App Router, SEO-Optimized) + FastAPI + PostgreSQL (RLS) + Local Windows Connector
Hosting Region	AWS Mumbai (`ap-south-1`) — Strict Indian Data Residency

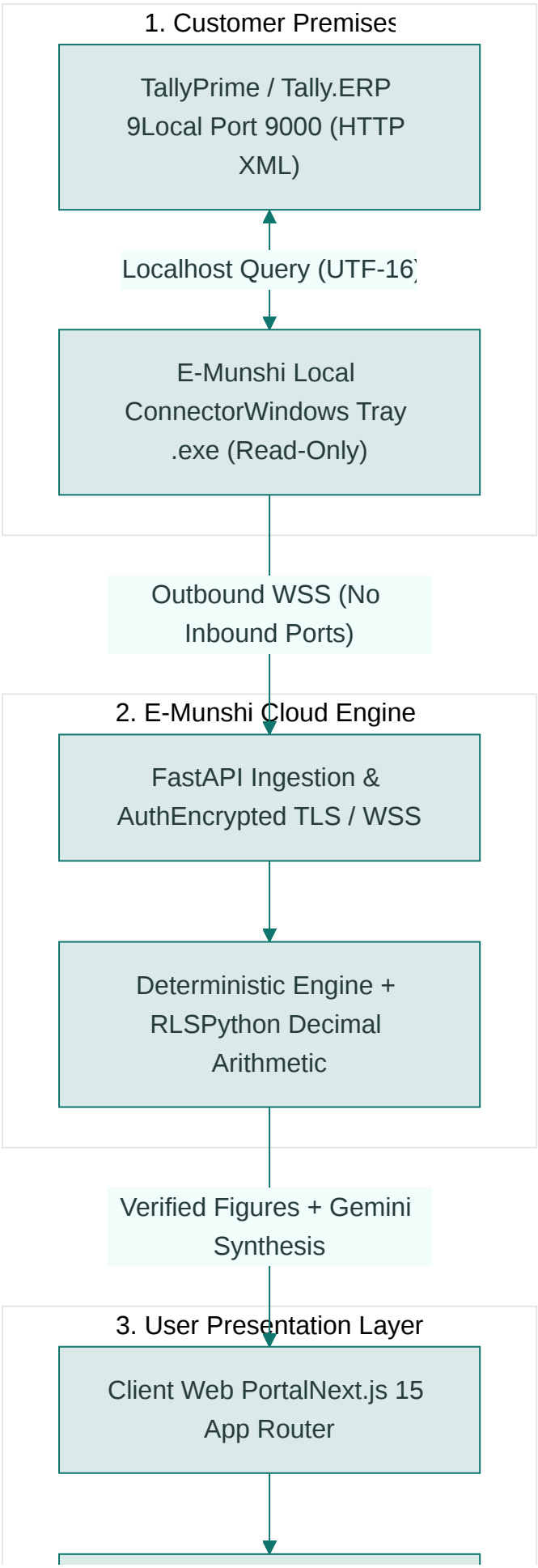
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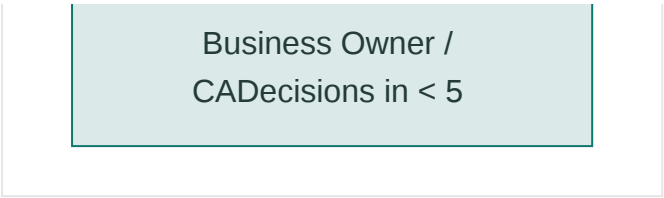
1.0 Executive Summary & Business Case

1.1 Product Purpose & Vision

E-Munshi (*"Munshi"* — the traditional, trusted financial advisor and bookkeeper in Indian commerce) is a high-security, cloud-native financial intelligence platform designed specifically for the **7+ million Indian enterprises** powered by Tally ERP (TallyPrime and Tally.ERP 9).

The platform bridges the critical **information accessibility gap** that exists between business leadership and their accounting ledgers. E-Munshi enables business owners, managing directors, and Chartered Accountants to query their financial records in plain English (and Hinglish) and receive **instantaneous, deterministic, and fully source-cited financial answers**—eliminating days of waiting for manual reports.


1. Customer Premises
TallyPrime / Tally.ERP 9Local Port 9000 (HTTP XML)
Localhost Query (UTF-16)
E-Munshi Local ConnectorWindows Tray .exe (Read-Only)
Outbound WSS (No Inbound Ports)
2. E-Munshi Cloud Engine
FastAPI Ingestion & AuthEncrypted TLS / WSS
Deterministic Engine + RLSPython Decimal Arithmetic
Verified Figures + Gemini Synthesis
3. User Presentation Layer
Client Web PortalNext.js 15 App Router



1.2 The Core Problem: The Financial Access Gap

Today, extracting basic operational answers from Tally—such as *"What is my actual disposable cash balance today?"* or *"Which debtors have overdue balances older than 60 days?"*—requires an arduous multi-step workflow:

1. Open Tally Prime on a specific designated office machine.
2. Navigate dense multi-tier accounting menus (Gateway of Tally → Display More Reports → Account Books...).
3. Configure custom date filters, period ranges, and bill-wise allocation toggles.
4. Export raw data tables to Microsoft Excel.
5. Manually cleanse and cross-tabulate figures in spreadsheets.
6. Transmit spreadsheets over unencrypted messaging channels (e.g., WhatsApp) to external accountants.
7. Endure an average turnaround latency of **24 to 72 hours** to receive a summarized figure.

This operational friction frequently results in:

- **Working Capital Mismanagement:** Leadership lacks instant visibility into true liquid cash reserves.
- **Delayed Collections:** Overdue receivables remain unchased because aging analysis is compiled only at month-end.
- **GST & Compliance Anxiety:** Discrepancies between physical ledgers and return filings are detected too late.
- **Strategic Inertia:** High-impact procurement and investment decisions are made on intuition rather than verified books.

1.3 The Solution: Instant, Verifiable Financial Intelligence

E-Munshi replaces this 7-step latency loop with a unified digital experience:

- **Query to Answer in Under 5 Seconds:** Leadership asks questions via a modern conversational interface or consults high-density interactive dashboards.
- **100% Deterministic Arithmetic:** Unlike conventional AI tools that estimate or synthesize figures, **E-Munshi hard-blocks LLM calculations**. Every figure is produced by verified SQL queries and Python `Decimal` routines running against raw accounting rows. The Large Language Model (Google Gemini 2.0 Flash) is strictly employed as an **interpreter and explainer**.
- **Audit-Grade Citations:** Every displayed metric carries a verifiable provenance badge indicating the exact source report, voucher timestamps, and company sync status.

1.4 Business & Market Justification

Strategic Dimension	Market Reality & E-Munshi Advantage
Addressable Market	Over 7,000,000 active Tally installations across India. The MSME digital enablement space represents an addressable market exceeding ₹3,500 Crores.
Validated Market Fit	Direct competitors such as Riko AI have established proven commercial willingness-to-pay among Indian SMEs and CA firms.
Capital Efficiency	E-Munshi's optimized micro-architecture maintains monthly cloud hosting expenses below ₹6,000 during the initial pilot phase, ensuring break-even with fewer than 15 active paid accounts.
Rapid Pilot Velocity	Designed for complete end-to-end production readiness within 30 calendar days, onboarding three validated pilot companies with zero downtime.
Commercial Model	Tiered software-as-a-service model starting at ₹3,999/year per enterprise, scaling to ₹18,000/year for multi-firm CA practice licenses.

2.0 Strategic Goals & Value Metrics

2.1 Core Business Objectives

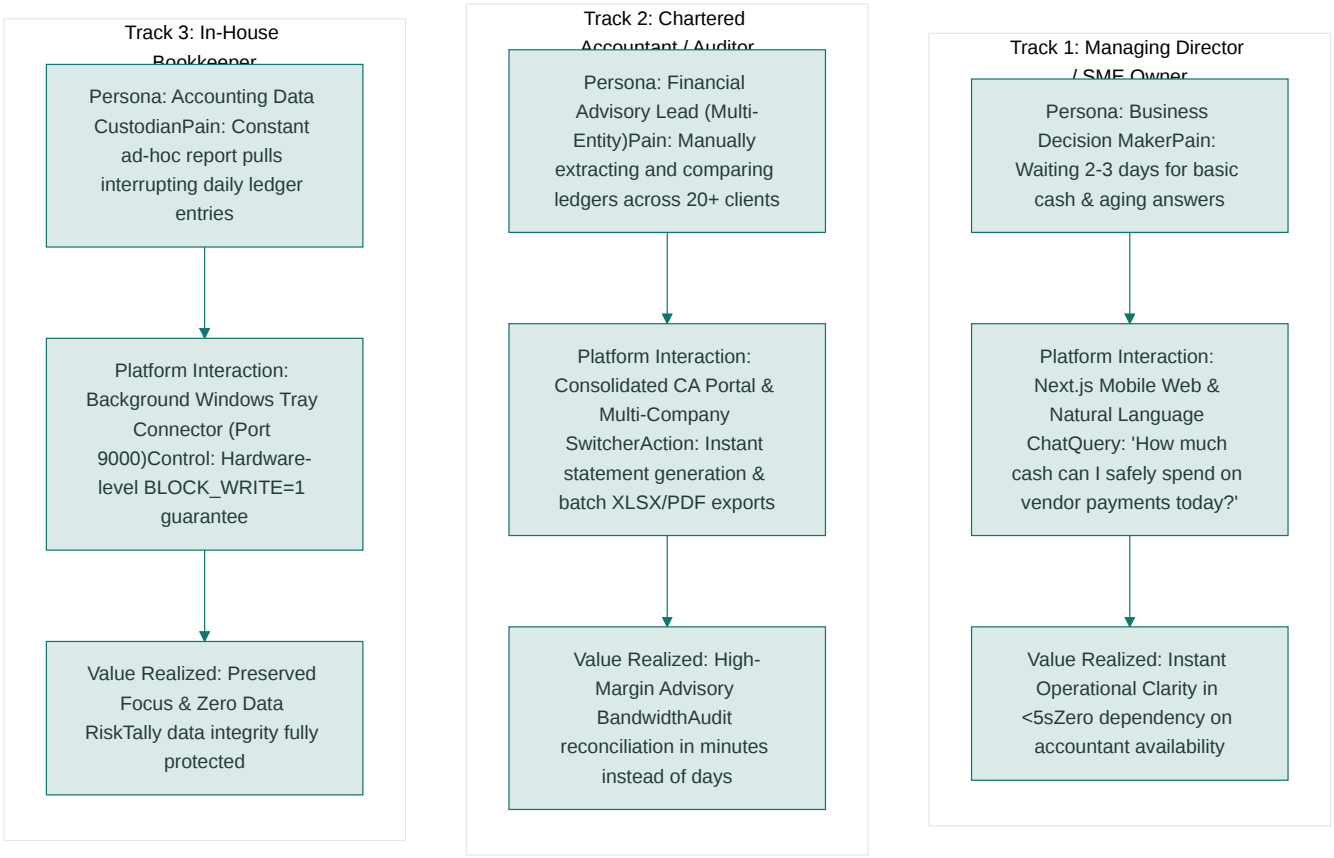
1. **Zero-Friction Deployment:** Enable any Indian enterprise with TallyPrime to pair their desktop connector and visualize verified dashboard figures within **10 minutes** without technical assistance.
2. **Defensible Financial Accuracy:** Guarantee a **100% computational match** between E-Munshi generated reports and native Tally statements across all tested periods.
3. **SEO-Driven Enterprise Acquisition:** Leverage a high-performance **Next.js 15 App Router** frontend to deploy server-rendered marketing landing pages, financial glossaries, and Tally utility tools that rank organically on search engines for high-intent accounting keywords.
4. **Data Sovereignty Assurance:** Adhere strictly to Indian regulatory mandates, ensuring all enterprise data persists and processes exclusively within the **AWS Mumbai (** `ap-south-1` **)** region.

2.2 Client-Centric Success KPIs

Metric Category	Target Value	Measurement Methodology
Computational Accuracy	**100.0%** match	Automated reconciliation of Trial Balance, P&L, and Bills Outstanding against Tally outputs across 3 consecutive closing periods.

Metric Category	Target Value	Measurement Methodology
End-to-End Query Latency	**< 3.0s** (P50) / **< 5.0s** (P95)	Time elapsed from user message submission to the rendering of the first streamed answer token over Server-Sent Events (SSE).
Initial Sync Duration	**< 5 minutes**	Total ingestion and encryption time for a 5,000-voucher enterprise dataset over standard 25 Mbps broadband.
Delta Sync Latency	**< 30 seconds**	Watermark-based synchronization cycle for up to 500 modified voucher records.
System Uptime	**≥ 99.5%**	Continuous external synthetic monitoring of web portal and API health endpoints.
Web Performance & SEO	**≥ 95/100** Core Web Vitals	Google Lighthouse performance and SEO benchmarks for all public marketing and educational routes.

3.0 Target User Personas & Client Workflows



3.1 Primary Personas

Persona A: The SME Business Owner / Managing Director

- **Context:** Operates a manufacturing, distribution, or service business generating ₹3 Cr to ₹50 Cr annually. Has limited formal accounting expertise.
- **Core Pain Point:** Needs real-time cash, margin, and receivable metrics during vendor negotiations or board meetings without waiting for internal accountants.
- **Primary Interface:** Next.js mobile-responsive web dashboard and conversational chat interface.

Persona B: The Practicing Chartered Accountant (CA)

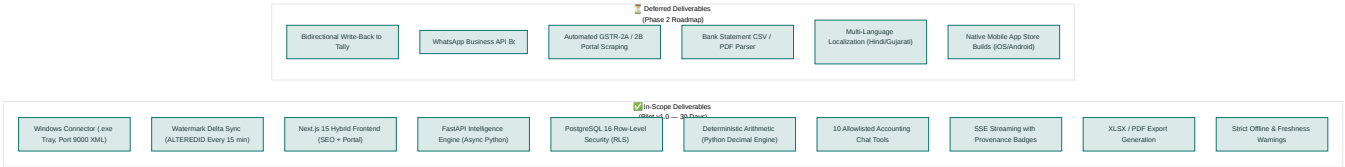
- **Context:** Manages 20 to 100 enterprise client accounts across distinct Tally installations.
- **Core Pain Point:** Loses dozens of billable hours each week pulling manual Trial Balances, cross-checking aging receivables, and answering routine status inquiries.
- **Primary Interface:** Multi-organization workspace switcher, side-by-side statement comparison, and batch export tools.

Persona C: The In-House Senior Bookkeeper / Accountant

- **Context:** Manages the primary Windows desktop hosting TallyPrime in the client's administrative office.
- **Core Pain Point:** Overwhelmed by daily repetitive requests for ledger statements and aging sheets from sales and leadership. Concerned about data security and accidental modification of books.
- **Primary Interface:** Read-only Windows system tray connector with explicit zero-write guarantees (`BLOCK_WRITE=1`).

4.0 Scope & Boundary Management

4.1 In-Scope Deliverables (Pilot Release Candidate — Day 30)



Component Area	Deliverable Details	Verification Criteria
F-01: Tally Connector	Single-executable 64-bit Windows tray service. Zero third-party runtime dependencies. Outbound TLS WebSocket only.	Successful install and handshake on Windows 10/11 without admin firewall adjustments.

Component Area	Deliverable Details	Verification Criteria
F-02: Watermark Delta Sync	Continuous extraction utilizing Tally's internal `ALTEREDID` state tracking every 15 minutes.	Incremental push transfers only updated rows; sync completes in <30 seconds.
F-03: Web Platform	Built on **Next.js 15 (App Router)**. Houses public SEO acquisition surfaces alongside the protected SaaS dashboard.	100% responsive across mobile, tablet, and desktop viewports; Lighthouse score ≥ 95.
F-04: Deterministic Engine	Python `Decimal` computations for Receivables Aging (0-30, 31-60, 61-90, 90+), P&L, Trial Balance, and Cash Flow.	Mathematical parity with Tally native statements across 100% of sample vouchers.
F-05: AI Intelligence Interface	Streaming conversational engine powered by Google Gemini 2.0 Flash via LiteLLM with 10 allowlisted accounting tools.	Explanations strictly reference deterministic outputs; includes full source provenance tags.
F-06: Enterprise Security	PostgreSQL 16 with Row-Level Security (RLS) enforced per organization, JWT session rotation, and automated audit logs.	Automated multi-tenant penetration tests verify zero cross-organization record leakage.

4.2 Explicitly Out-of-Scope (Deferred to Phase 2)

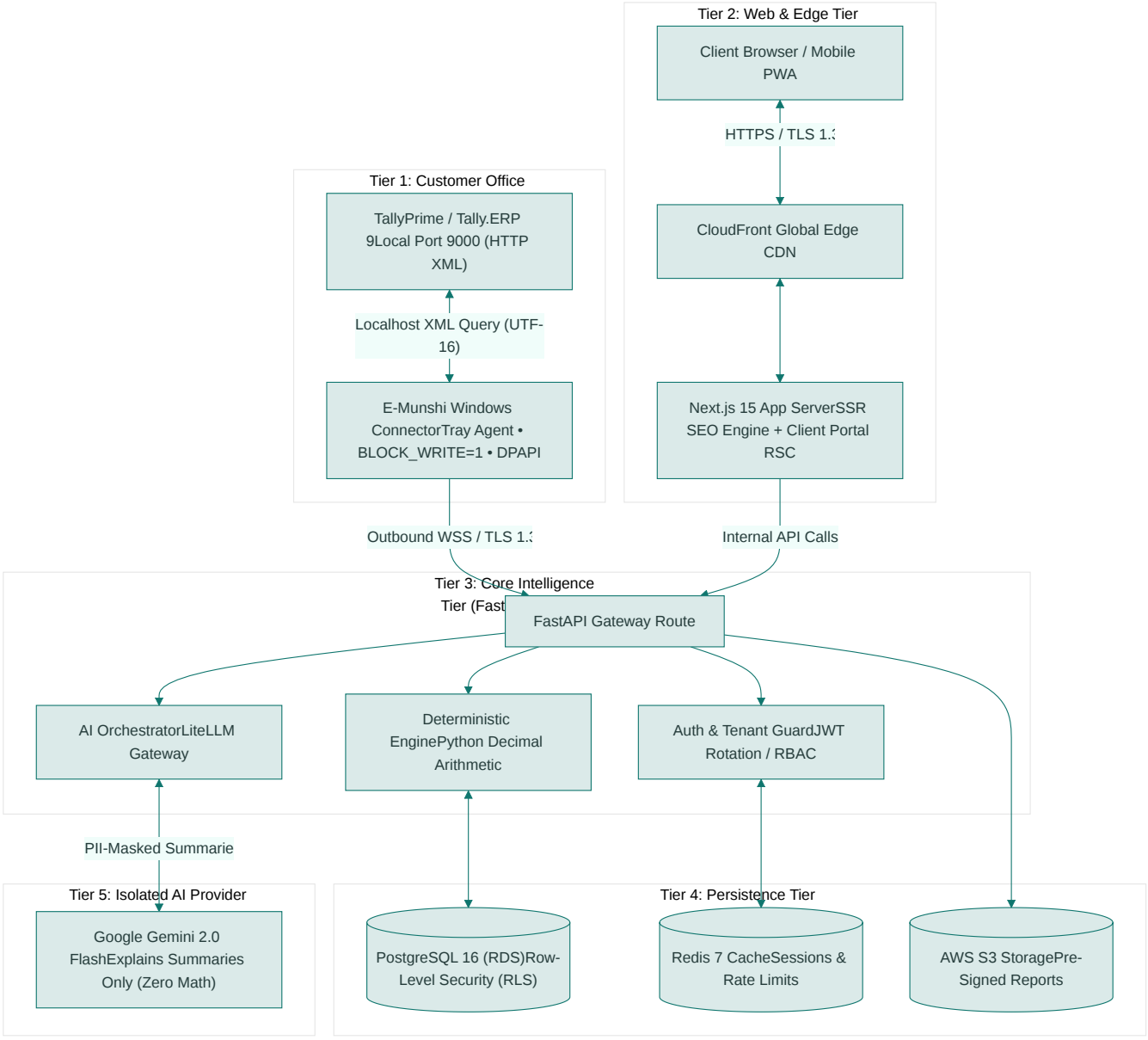
- **Bidirectional Write-Back to Tally:** Creating or modifying vouchers, journals, or ledgers in Tally is strictly disabled in v1.0. This prevents any risk of ledger corruption or compliance exposure during pilot rollout.
- **WhatsApp Conversational Bot:** Excluded from the 30-day pilot to prevent dependency on Meta Business verification lead times (typically 2 to 4 weeks).
- **Automated GSTN Portal Scraping:** Reconciliation against GSTR-2A/2B requires external government portal session handling, deferred to Phase 2.
- **Native App Store Distribution:** Responsive mobile PWA on Next.js satisfies mobile access requirements without App Store review overhead.

4.3 Explicit System Non-Goals

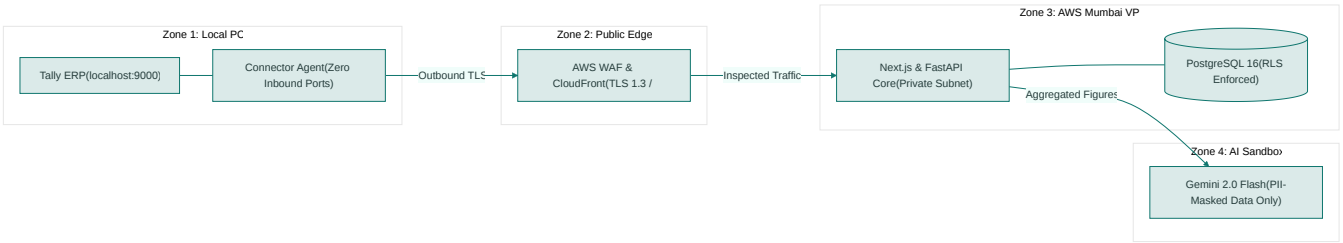
- E-Munshi **does not replace Tally ERP**. It operates strictly as a companion analytics and intelligence layer.
- E-Munshi **does not provide automated tax or legal counsel**. All outputs are provided for executive business intelligence; formal audit decisions remain with certified accountants.
- E-Munshi **does not ingest or retain unencrypted Tally binary data files (`.900` / `.tsf`)**. It stores only structured, normalized accounting rows extracted via official XML interfaces.

5.0 System Architecture & Infrastructure

5.1 End-to-End System Architecture



5.2 Network Security & Trust Boundaries



Trust Zone	Boundary Protection & Posture
Zone 1: Client Premises	The local connector executes within the user's desktop environment. It creates **zero listening network sockets**, permanently eliminating inbound attack surfaces. Communication with Tally is restricted to `localhost:9000`.
Zone 2: Transit Network	All transmission across public networks utilizes **TLS 1.3 encryption with HSTS enforcement**. WebSockets leverage certificate pinning and temporary rotating connection tokens.
Zone 3: Core Cloud VPC	Services run in private subnets within the AWS Mumbai region. PostgreSQL databases are inaccessible from the public internet. Access is strictly gated via application-level connection pooling and PostgreSQL Row-Level Security.
Zone 4: LLM Boundary	The AI model never receives raw database rows, bank account numbers, PANs, or bulk customer lists. Only aggregated, pre-calculated numerical summaries are transmitted for conversational framing.

6.0 Web Architecture & SEO Strategy (Next.js)

6.1 Architectural Rationale for Next.js 15

To satisfy both **rapid organic customer acquisition** and a **frictionless client dashboard experience**, E-Munshi utilizes **Next.js 15 (App Router)** as its primary web tier.

A traditional single-page application (SPA) built with pure client-side React suffers from poor search crawler indexability, sluggish Core Web Vitals on mobile cellular networks, and delayed initial loads. Next.js 15 resolves these constraints by providing a unified, hybrid rendering architecture:



6.2 Organic Acquisition & Technical SEO Engine

E-Munshi will capture high-intent accounting and business management queries organically by deploying programmatically generated, server-rendered content clusters:

SEO Content Cluster	Target Search Queries	Next.js Implementation Strategy
Tally Error & Report Guides	*"How to check overdue debtors in TallyPrime"*, *"Tally balance sheet difference troubleshooting"*	**Static Site Generation (SSG)** with pre-rendered markdown guides, generating 200+ high-ranking educational landing pages.
Interactive Financial Calculators	*"Working capital cycle calculator India"*, *"GST input tax credit calculator"*	**Hybrid SSR/Client Components** providing immediate calculation utility with prominent conversion banners to install the E-Munshi connector.
Persona-Specific Landing Pages	*"Tally for Chartered Accountants"*, *"Mobile Tally dashboard for business owners"*	**Optimized SSG Pages** equipped with dynamic OpenGraph meta-tags, schema.org JSON-LD microdata, and localized client social proof.

6.3 Core Web Vitals & Performance Budget

- **Largest Contentful Paint (LCP):** < 1.2 seconds (achieved via Next.js image optimization and edge caching).
- **Interaction to Next Paint (INP):** < 80 milliseconds (lightweight client bundles, selective hydration).
- **Cumulative Layout Shift (CLS):** 0.00 (rigid layout sizing and server-rendered fonts).
- **Structured Data:** Automated injection of `Organization`, `SoftwareApplication`, `FAQPage`, and `BreadcrumbList` schema objects into every public response header.

7.0 Tally ERP Integration & Sync Engine

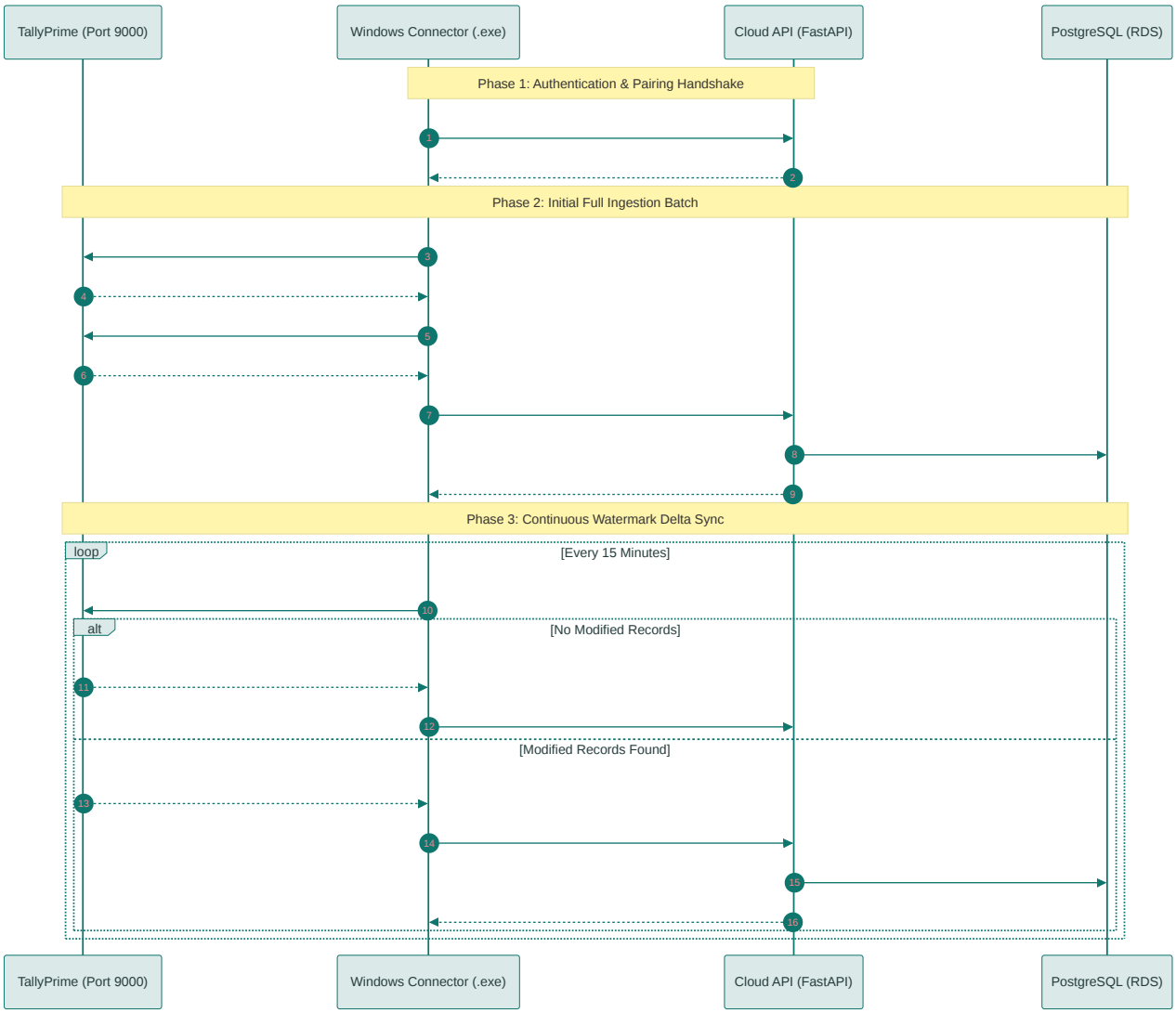
7.1 Protocol & Port Specifications

E-Munshi connects to TallyPrime and Tally.ERP 9 using Tally's official, documented **XML HTTP Server interface on port 9000**. This approach ensures complete compatibility without requiring proprietary DLL injection, database cracking, or unsupported binary modifications.

```
Request Payload:
POST http://localhost:9000/
Content-Type: text/xml; charset=utf-16
Body: TDL (Tally Definition Language) XML Request Envelope

Response Payload:
HTTP/1.1 200 OK
Content-Type: text/xml; charset=utf-16le
Body: Structured XML Response Envelope containing Ledger/Voucher collection
```

7.2 Data Ingestion Lifecycle



7.3 Data Freshness & Offline Degradation Policy

Metric / Report Category	Refresh Mechanism	Staleness Threshold	UI Freshness Presentation
Cash & Liquid Balances	Delta Sync (15 min)	30 minutes	`● Live (Synced 12 mins ago)`
Receivables Aging Analysis	Delta Sync (15 min)	2 hours	`● Verified (As of 14:30 today)`
P&L / Statement of Operations	On-Demand Live Query	5 minutes	`● Real-Time Tally Pull`
Chart of Accounts (Masters)	Incremental Check	24 hours	`● System Configured`

> [!IMPORTANT] **Graceful Offline Degradation:** If the host machine running Tally is closed or disconnected from the internet, E-Munshi **never displays blank screens**

> **or silent stale data**. The interface prominently displays an amber alert: `⚠ Tally Connector Offline since [Time] — Displaying verified records as of [Timestamp]`.

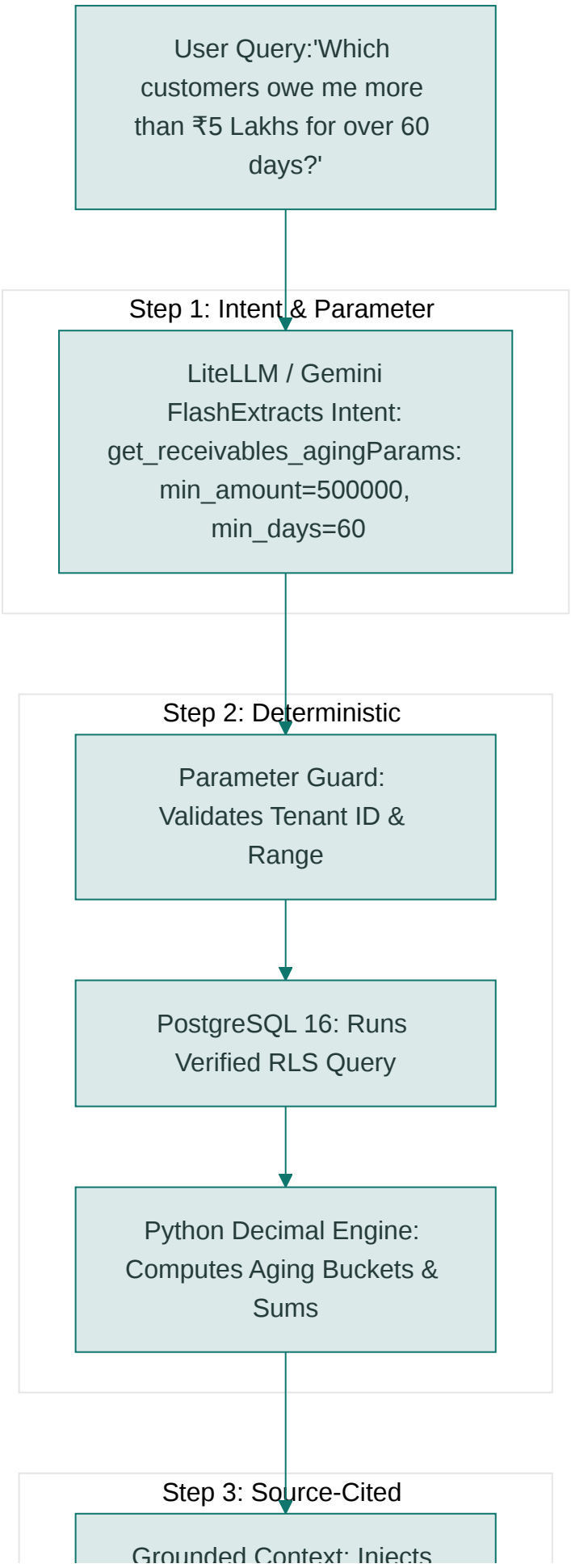
8.0 AI Orchestration & Deterministic Financial Engine

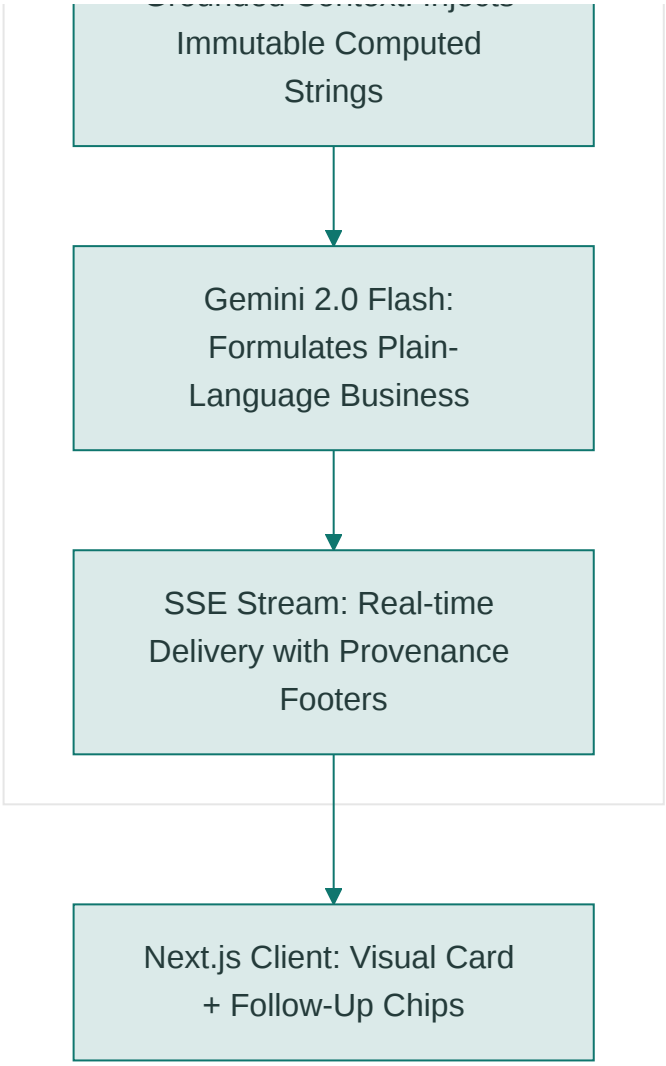
8.1 The Explainer-Not-Calculator Principle

A fundamental design requirement for E-Munshi is the **elimination of artificial intelligence hallucinations in financial mathematics**. Large Language Models are probabilistic token predictors and are fundamentally incapable of guaranteeing arithmetic precision.

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║   The Large Language Model is strictly an EXPLAINER, never a CALCULATOR.
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║   • All sums, balances, aging buckets, and ratios are calculated by
deterministic║
║     SQL queries and Python Decimal routines with fixed 2-place precision.
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║   • The LLM receives pre-calculated, verified numerical outputs as fixed
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║     immutable string values.
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║   • The LLM translates structured results into natural business language and
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║     attaches traceable audit citations.
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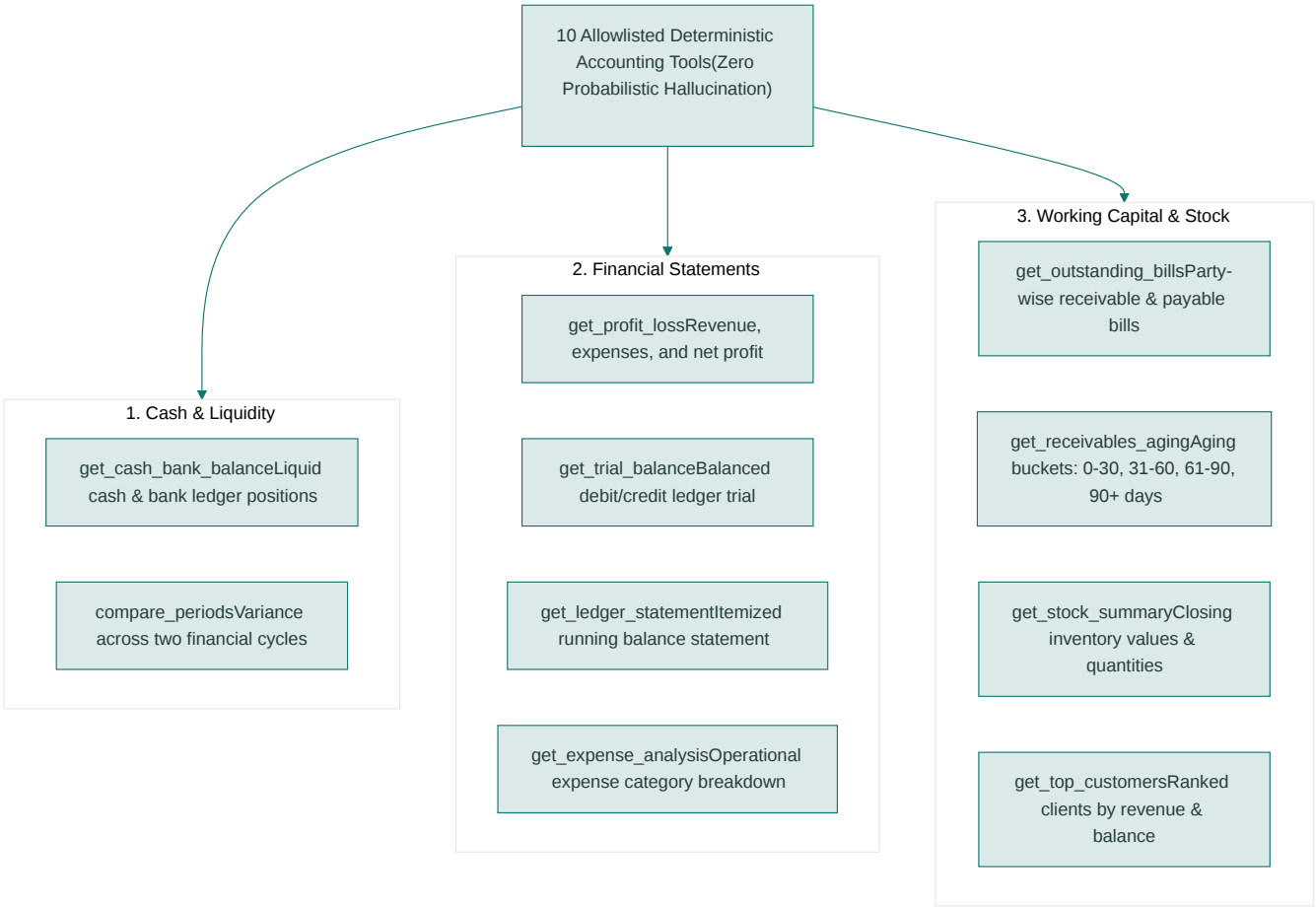
8.2 End-to-End Query Execution Pipeline


User Query:'Which customers owe me more than ₹5 Lakhs for over 60 days?'
Step 1: Intent & Parameter
LiteLLM / Gemini FlashExtracts Intent: get_receivables_agingParams: min_amount=500000, min_days=60
Step 2: Deterministic
Parameter Guard: Validates Tenant ID & Range
PostgreSQL 16: Runs Verified RLS Query
Python Decimal Engine: Computes Aging Buckets & Sums
Step 3: Source-Cited
Grounded Context: Injects



8.3 Allowlisted Financial Tools (Pilot Scope)

The orchestrator supports strictly ten allowlisted tool functions. Any query outside this allowlist is routed to a structured educational refusal.

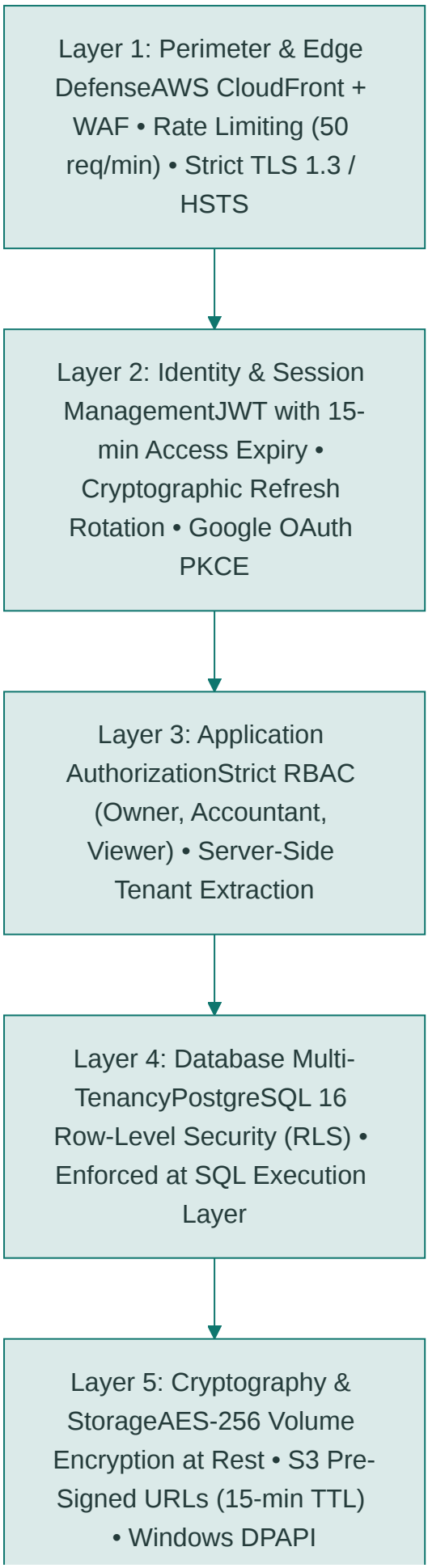


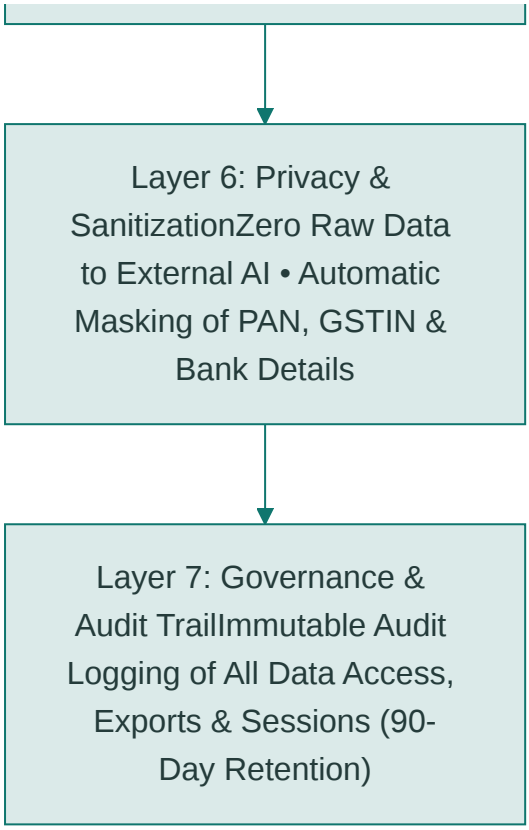
8.4 Anti-Hallucination & Numerical Guardrails

1. **String-Locked Currency Values:** Numerical values are converted to currency-formatted strings (e.g., `"₹14,25,000.00"`) prior to prompt injection, preventing the LLM from attempting scientific rounding or truncation.
2. **Backend Numerical Re-Verification:** Before the final streamed response reaches the client browser, an automated regex inspector cross-verifies all currency tokens against the deterministic payload. If an unregistered number is detected, the response is suppressed and replaced with raw structured tables.
3. **Prompt Injection Hardening:** Ledger names, voucher narrations, and party names are sanitized and wrapped in CDATA delimiter blocks before classification, preventing malicious Tally account names from overriding system instructions.

9.0 Security, Governance & Regulatory Compliance

9.1 Multi-Layer Defense Architecture


Layer 1: Perimeter & Edge DefenseAWS CloudFront + WAF • Rate Limiting (50 req/min) • Strict TLS 1.3 / HSTS
Layer 2: Identity & Session ManagementJWT with 15-min Access Expiry • Cryptographic Refresh Rotation • Google OAuth PKCE
Layer 3: Application AuthorizationStrict RBAC (Owner, Accountant, Viewer) • Server-Side Tenant Extraction
Layer 4: Database Multi-TenancyPostgreSQL 16 Row-Level Security (RLS) • Enforced at SQL Execution Layer
Layer 5: Cryptography & StorageAES-256 Volume Encryption at Rest • S3 Pre-Signed URLs (15-min TTL) • Windows DPAPI



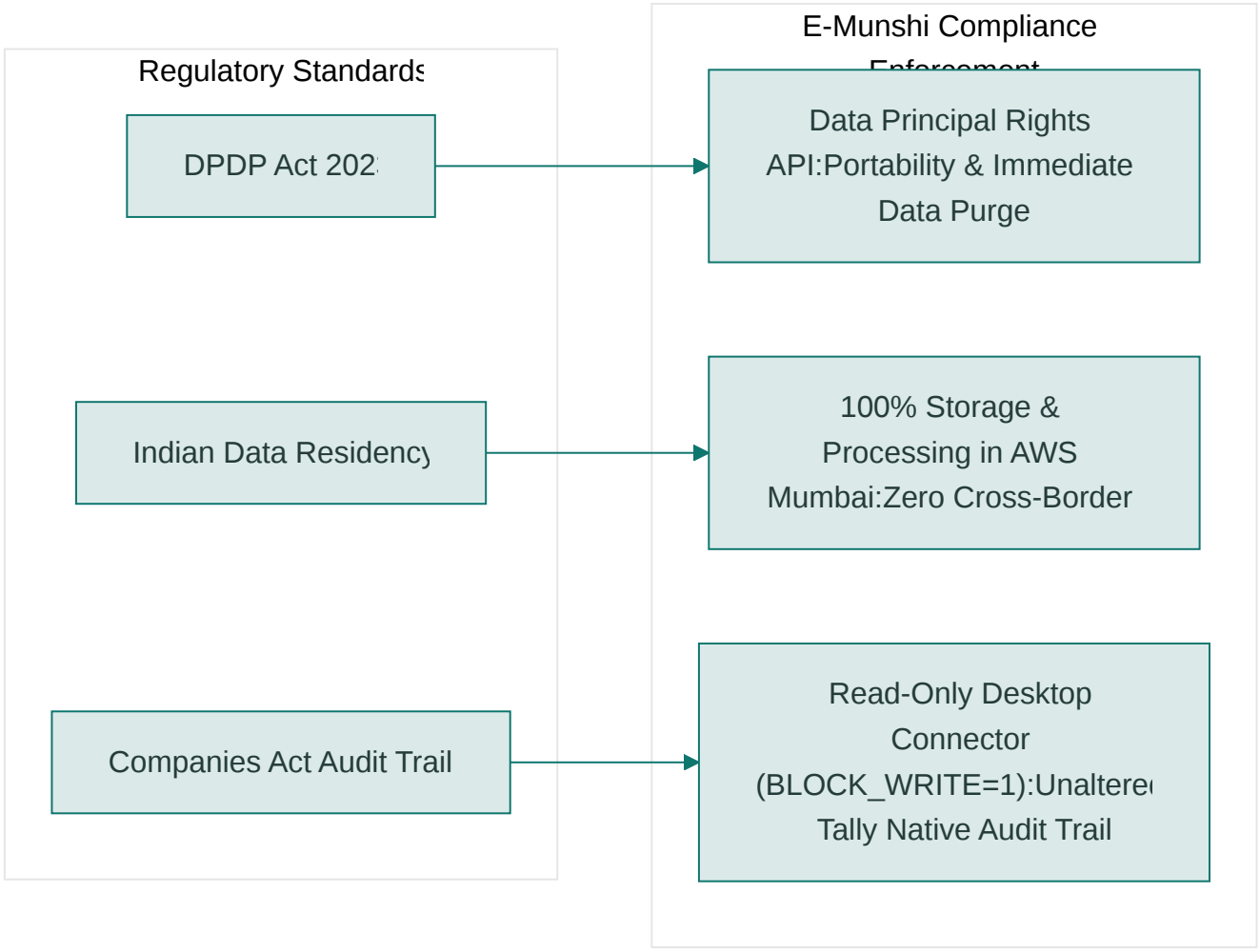
9.2 Multi-Tenant Data Isolation (PostgreSQL RLS)

Tenant segregation is enforced at the database engine level using PostgreSQL Row-Level Security. Application code cannot bypass organization isolation even in the event of an SQL injection attempt.

```
-- Enforcement of organization data segregation
ALTER TABLE vouchers ENABLE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation_policy ON vouchers
    FOR ALL
    USING (organization_id = NULLIF(current_setting('app.current_tenant_id',
true), '')::uuid);
```

9.3 Indian Regulatory Compliance & Data Residency



10.0 Technology Stack Specifications

10.1 Complete Technology Matrix

System Layer	Selected Technology	Version / Spec	Engineering Rationale
Web Tier & SEO	**Next.js (App Router)**	`15.x`	Hybrid architecture delivering lightning-fast static marketing pages for search indexing combined with dynamic server-rendered client applications.
Frontend Runtime	**React / TypeScript**	`19.x` / `5.x`	Type-safe enterprise code quality; React Server Components for zero-bundle table rendering.
Styling & Design System	**Tailwind CSS & Radix UI**	Modern Utilities	Clean, accessible design system ensuring high contrast, clean typography, and responsive layouts.

System Layer	Selected Technology	Version / Spec	Engineering Rationale
Data Visualization	**Recharts**	`2.x`	Hardware-accelerated SVG charts optimized for financial time series and aging distribution.
Core API Backend	**FastAPI (Python)**	`3.12` / `0.115+`	Async native event loop, Pydantic v2 strict typing, seamless integration with financial math libraries.
Database Engine	**PostgreSQL**	`16.x`	ACID transactions, native Row-Level Security, JSONB indexing for flexible TDL attributes.
Caching & Ephemeral Store	**Redis**	`7.x`	Sub-millisecond session validation, rate-limiting counters, and sync watermark queues.
AI LLM Gateway	**Google Gemini 2.0 Flash via LiteLLM**	Production API	High processing speed, large context handling, economical cost profile, and strong Hinglish language comprehension.
Authentication Service	**Supabase Auth / Custom JWT**	Industry Standard	Secure OAuth 2.0 PKCE, multi-factor support, localized India hosting compliance.
Desktop Connector	**Node.js (TypeScript) Packaged EXE**	Windows 64-bit	Packaged as a single standalone executable using `pkg` / Windows DPAPI. Zero runtime setup for client.
Object Storage	**AWS S3**	Mumbai (`ap-south-1`)	Pre-signed private URLs for scheduled Excel and PDF exports; strict lifecycle rules.
Observability & Error Tracking	**Sentry & Grafana Cloud**	Cloud Hosted	Real-time frontend exception monitoring, Core Web Vitals tracking, and API latency dashboards.

11.0 Functional Requirements

11.1 FR-1: Tally Local Connector Agent

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-1.1	Standalone zero-dependency Windows desktop installation.	**P0**	Installs cleanly on standard Windows 10/11 64-bit systems without requiring manual installation of Node.js, Python, or .NET runtimes.

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-1.2	System tray presence with dynamic connection health badges.	**P0**	Tray icon displays clear operational states: 🟢 Online & Synced, 🟡 Delta Syncing, 🔴 Tally Offline.
FR-1.3	Communication over standard localhost XML interface.	**P0**	Communicates with Tally on `http://localhost:9000` using standard TDL envelopes; compatible with TallyPrime 2.x–5.x and Tally.ERP 9.
FR-1.4	Purely outbound network architecture.	**P0**	Network inspection confirms zero listening ports opened by the connector; establishes outbound TLS WebSocket to the cloud backend.
FR-1.5	Initial sync extraction capabilities.	**P0**	Successfully extracts Company Profile, Chart of Accounts, current financial year Vouchers, and Bill Allocations within 5 minutes for 5,000 vouchers.
FR-1.6	Incremental watermark synchronization.	**P0**	Queries Tally using `ALTEREDID > watermark` every 15 minutes; transmits only modified rows.
FR-1.7	Hard-coded write prohibition (`BLOCK_WRITE=1`).	**P0**	Code-level compilation guarantee blocking any HTTP XML write actions to Tally. Any write attempt throws an alert and terminates.
FR-1.8	One-time pairing code handshake.	**P0**	Connectors pair via a 6-digit cryptographic pairing code with a 10-minute expiry; single-use validation enforced by Redis.

11.2 FR-2: Identity, Access & Organization Management

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-2.1	Multi-tenant organization boundaries.	**P0**	Users can create or belong to multiple business organizations; data is strictly segregated per organization ID.
FR-2.2	Google OAuth and Email/Password authentication.	**P0**	Secure login utilizing PKCE flow; session tokens issued with 15-minute access lifespan and 7-day sliding refresh.
FR-2.3	Granular Role-Based Access Control (RBAC).	**P0**	Supports three distinct organizational roles: `OWNER` (full access), `ACCOUNTANT` (reporting and chat), and `VIEWER` (read-only dashboard).

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-2.4	Multi-entity company switching.	**P0**	Users with multiple Tally companies can switch contexts seamlessly via an app header dropdown; refreshes metrics instantly.

11.3 FR-3: Financial Overview Dashboard

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-3.1	Liquid Cash & Bank Position card.	**P0**	Displays aggregated cash and bank ledger balances; exactly matches Tally Balance Sheet Cash/Bank Group.
FR-3.2	Sales Performance & Month-over-Month variance.	**P0**	Displays current month sales alongside previous month total and percentage change; reconciles with Tally Sales Accounts total.
FR-3.3	Top Debtors exposure ranking.	**P0**	Visualizes the top 5 debtors with greatest total outstanding balance; verifies names and values against Bills Outstanding.
FR-3.4	Receivables Aging distribution breakdown.	**P0**	Summarizes overdue balances across standardized aging buckets: 0–30, 31–60, 61–90, and 90+ days.
FR-3.5	Prominent data freshness indicator.	**P0**	Every dashboard card displays a sync timestamp (e.g., `"As of 14:15 today"`); displays amber warning banner if connector is offline >30 min.

11.4 FR-4: Natural Language Financial Intelligence Interface

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-4.1	Plain English & Hinglish conversational processing.	**P0**	Accurately interprets natural language queries (e.g., *"What is my cash balance?"* or *"Mera total overdue kitna hai?"*).
FR-4.2	Audit-grade provenance citation.	**P0**	Every AI response terminates with a structured footer: `"Source: [Tally Report Name] · Synced: [Timestamp] · Company: [Name]"`.
FR-4.3	Server-Sent Events (SSE) progressive token streaming.	**P0**	Renders intermediate thought stages (*"Fetching ledger..."* → *"Computing aging..."*) and streams the synthesized explanation in real time.

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-4.4	Controlled out-of-scope refusals.	**P0**	Queries outside the 10 allowlisted accounting tools return polite, structured suggestions without hallucinating speculative answers.
FR-4.5	Contextual follow-up suggestions.	**P1**	Automatically generates 2–3 clickable chip suggestions following each answer (e.g., *[Show 90+ Day Debtors]*, *[Export to Excel]*).

11.5 FR-5: Core Financial Reporting & Exports

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-5.1	Trial Balance generation with dynamic date selection.	**P0**	Displays complete hierarchy of debit and credit ledgers; debit and credit totals match Tally to the exact paise.
FR-5.2	Profit & Loss Statement view.	**P0**	Formats revenue, direct costs, and indirect expenses; verified net profit matches native Tally P&L.
FR-5.3	Itemized Ledger Account Statement.	**P0**	Generates detailed statement for any selected ledger including opening balance, chronologically ordered entries, and running closing balance.
FR-5.4	Excel (XLSX) and PDF report exports.	**P0**	Single-click export generation; downloads branded spreadsheet matching screen figures via secure 15-minute S3 pre-signed URL.

11.6 FR-6: Connector Lifecycle & Diagnostic Management

Requirement ID	Specification Details	Priority	Acceptance Criteria
FR-6.1	Real-time connector online/offline monitoring.	**P0**	Cloud portal displays active heartbeat status; notifies user if heartbeats lapse for more than 5 minutes.
FR-6.2	Diagnostic logs & sync history inspector.	**P1**	Displays previous 20 sync runs, row counts, transfer duration, and error codes to assist in customer support diagnostics.

12.0 Non-Functional Requirements

12.1 Performance Targets



- **Dashboard Initial Load:** Under 1.5 seconds on standard 4G broadband connections.
- **AI Conversational Latency:** First streamed token rendered within 3.0 seconds (P50) and under 5.0 seconds (P95).
- **Statement Compilation Latency:** Trial Balance and P&L statements compiled in under 2.0 seconds for enterprise entities holding up to 100,000 vouchers.
- **API Route Response Time:** Non-AI transactional endpoints respond in under 300 milliseconds (P95).

12.2 Availability, Reliability & Disaster Recovery

- **Service Availability SLA:** 99.5% operational availability for web portal and API services during business hours (08:00–22:00 IST).
- **Database Backup Schedule:** Automated full daily snapshots with continuous point-in-time recovery (PITR) retained for 7 days.
- **Disaster Recovery RTO & RPO:** Recovery Time Objective (RTO) < 4 hours; Recovery Point Objective (RPO) < 15 minutes (bounded by the delta sync cycle).
- **Connector Network Resilience:** Automatic exponential reconnection backoff (5s, 15s, 30s, 60s) upon local network drops; resumes from existing watermark with zero record duplication.

12.3 Scalability Envelope (Pilot Phase Targets)

- Support up to **50 concurrent enterprise organizations**.
- Support up to **20 distinct Tally companies** managed under a single CA firm account.
- Ingestion engine architected to handle up to **500,000 vouchers** per enterprise entity without memory degradation.
- Cloud API architected to handle up to **10,000 conversational intelligence queries** daily.

13.0 User Stories & Acceptance Criteria

Epic 1: Effortless Connector Setup & Sync

- **User Story US-1.1:** *As an SME business owner, I want to download and pair the E-Munshi Windows connector in under 3 minutes, so that my Tally data is accessible without complex IT configuration.*
  - **Given:** The user is logged into the Next.js web portal on an authenticated account, and TallyPrime is open on their Windows PC.
  - **When:** The user downloads the `.exe` installer, launches it, and submits the 6-digit pairing code displayed on the portal.

- **Then:** The connector validates credentials over outbound TLS WebSocket, completes master ledger extraction, and displays a green 🟢 status badge within 5 minutes.
- **And:** No inbound firewall ports or router port-forwarding rules are required.

Epic 2: Real-Time Plain-Language Financial Inquiries

- **User Story US-2.1:** *As a Managing Director, I want to ask "Who owes me the most money and for how long?" in natural English, so that I can immediately direct my collection team.*
  - **Given:** The enterprise has synchronized current bill-wise outstanding vouchers from Tally.
  - **When:** The user submits the question into the conversational interface.
  - **Then:** The system executes deterministic receivables queries, groups debtors by aging buckets, and presents a ranked list with exact currency values.
  - **And:** The answer includes a data source provenance badge indicating sync time and company name.

Epic 3: Multi-Entity Switching for Chartered Accountants

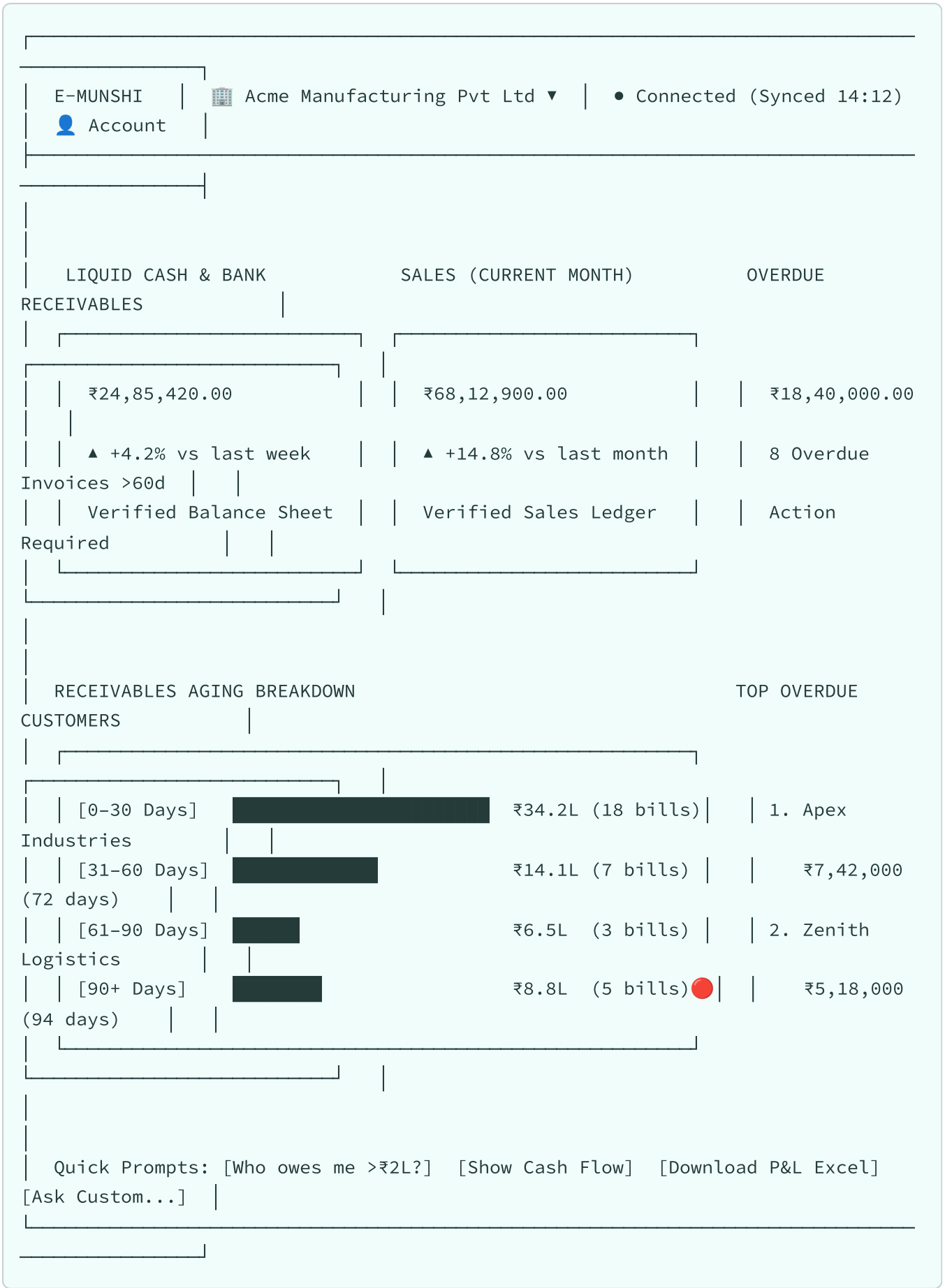
- **User Story US-3.1:** *As a practicing CA managing 15 clients, I want to switch between client books via a top dropdown, so that I can advise clients without logging into multiple accounts.*
  - **Given:** The CA user holds authorized `ACCOUNTANT` or `OWNER` permissions across multiple organizations.
  - **When:** The CA selects a different entity from the organization dropdown.
  - **Then:** Dashboard metrics, AI chat session context, and report views instantly refresh to the selected client.
  - **And:** Strict PostgreSQL RLS guarantees that no data from unauthorized client entities can be leaked or queried.

14.0 UX/UI Specifications & Wireframe Layouts

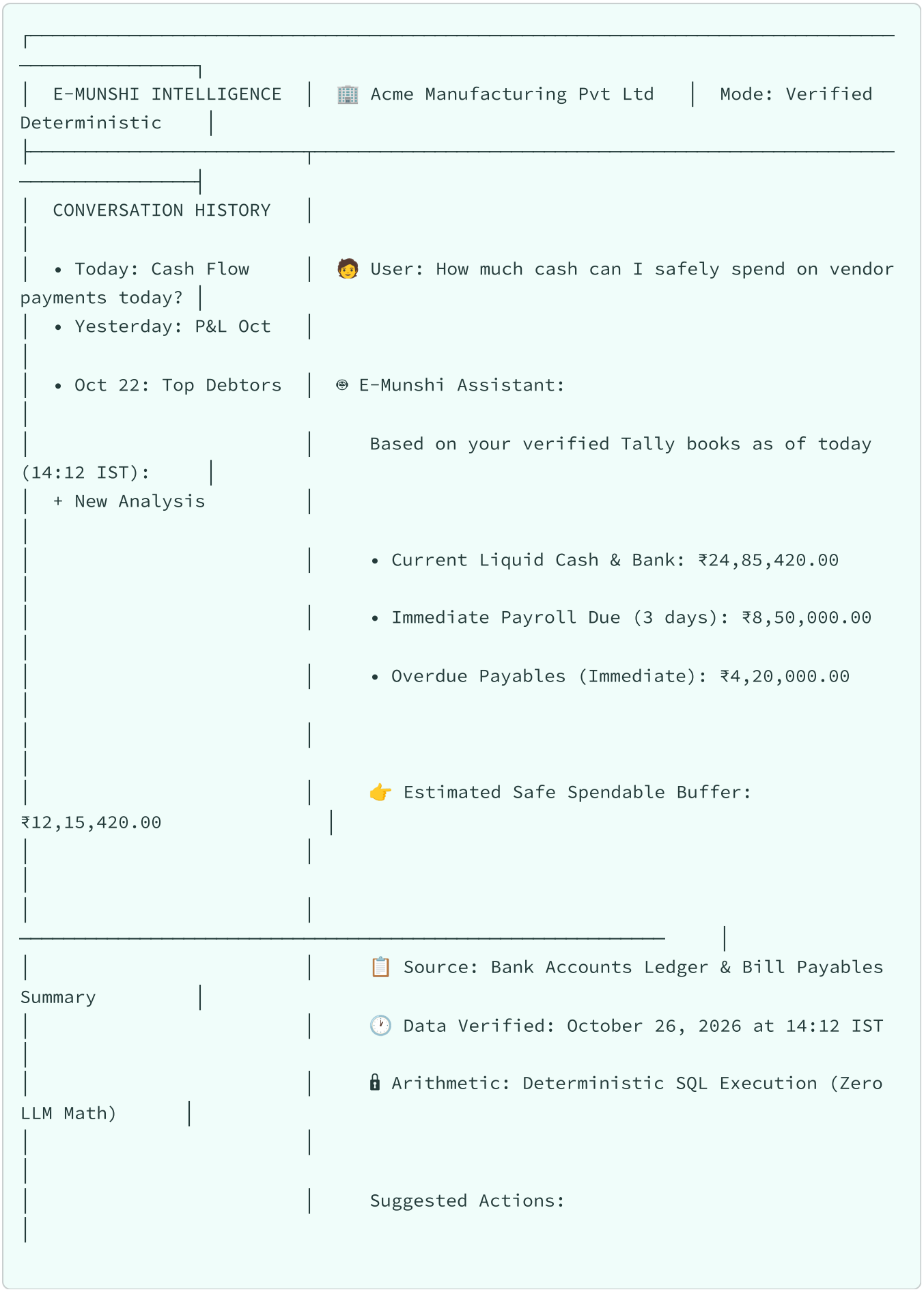
14.1 Design Philosophy & Human Factors

1. **Executive Scannability:** Key financial health metrics must be understandable within 3 seconds of viewing.
2. **Uncompromising Data Provenance:** Every metric must explicitly show its data origin and sync timestamp.
3. **Responsive Hybrid Presentation:** Seamless navigation across mobile smartphones, iPads, and multi-monitor desktop environments.

14.2 Wireframe 1: Executive Financial Dashboard (Next.js Portal)



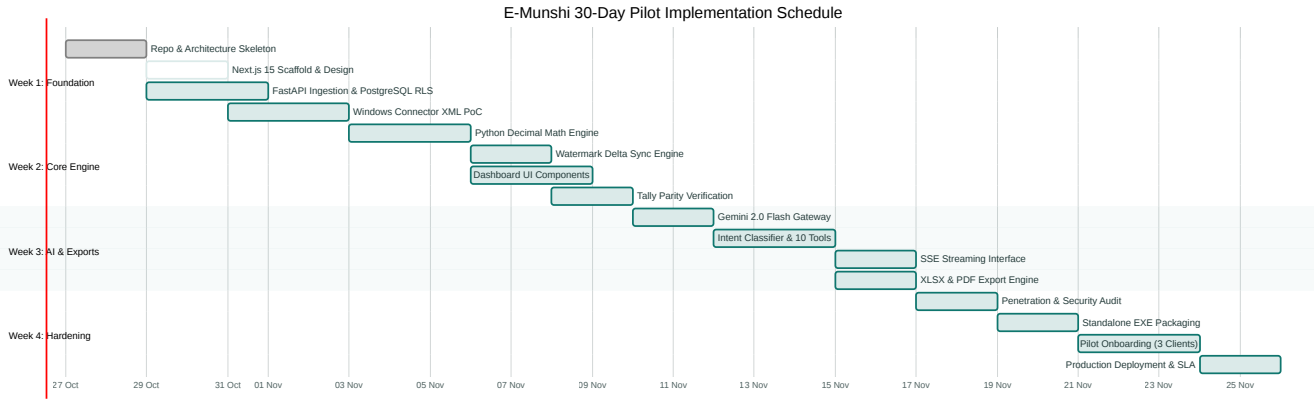
14.3 Wireframe 2: AI Financial Intelligence Chat & Source Citations



```
│                          │     [View Bank-Wise Balances]  [List Overdue
Payables] [Export MIS] │
│                          │
│
│                          │
┌──────────────────────────────────────────────────────────────────┐     │
│                          │  │ Ask a financial question about your Tally
books...     [➤] │     │
│                          │
└──────────────────────────────────────────────────────────────────┘     │
└──────────────────────────┴──────────────────────────────────────────────
──────────────┘
```

15.0 Implementation Roadmap & Milestone Schedule

15.1 30-Day Sprint Schedule



15.2 Quality Gates & Acceptance Milestones

Gate ID	Timeline	Gate Deliverable & Verification Criteria
M1	**Day 4**	**Tally Local Extraction Proof-of-Concept:** Connector successfully extracts Chart of Accounts, Trial Balance, and Bills Outstanding from live TallyPrime on `localhost:9000`.
M2	**Day 7**	**Secure Cloud Ingestion:** End-to-end pipeline functional: Connector pairs via one-time token, encrypts data, and persists records in PostgreSQL with active RLS tenant isolation.
M3	**Day 14**	**Deterministic Financial Engine Verification:** Dashboard metrics match Tally native reports to 100% accuracy across 3 distinct test accounting periods.

Gate ID	Timeline	Gate Deliverable & Verification Criteria
M4	**Day 21**	**Conversational Intelligence Operational:** 10 allowlisted accounting query types executed successfully via SSE streaming with complete source provenance badges.
M5	**Day 25**	**Security & Isolation Sign-Off:** Automated multi-tenant penetration tests confirm zero cross-tenant leakage. Strict rate-limiting active on all routes.
M6	**Day 30**	**Production Pilot Launch:** 3 verified pilot enterprises operating with continuous delta sync and zero critical data defects.

16.0 Commercial Model & Operating Budget

16.1 Pilot Infrastructure & LLM Budget (AWS Mumbai)

Infrastructure Service	Specific Provider / Instance	Monthly Cost (INR)
API & Web Server	AWS Lightsail / ECS (2 vCPU, 4GB RAM, Mumbai)	₹1,650
Managed Relational DB	Supabase Pro / AWS RDS PostgreSQL (Mumbai)	₹2,100
In-Memory Cache	Upstash Redis (Serverless Tier)	₹350
Frontend CDN & Edge	AWS CloudFront + S3 Origin (Mumbai)	₹200
LLM Inference Engine	Google Gemini 2.0 Flash (~20,000 queries/month)	₹1,200
Object File Storage	AWS S3 Standard (Report Exports)	₹150
Domain & DNS Security	Cloudflare Pro / Registry	₹120
Total Monthly Burn	**Continuous Production Hosting**	**₹5,770 / month**

16.2 Three-Month Operational Cost Model

- **Total 3-Month Cloud Infrastructure:** ₹17,310
- **Domain, Security & Communications Infrastructure:** ₹2,500
- **AI Tooling, Development & CI/CD Pipelines:** ₹7,500
- **Total 90-Day Expenditure Envelope: ₹27,310**

> [!TIP] **Capital Efficiency Guarantee:** With total initial pilot expenses capped under **₹30,000**, the project achieves self-sustaining commercial profitability with fewer

> than **8 annual business subscriptions**.

16.3 Client Pricing & Monetization Structure

Commercial Subscription Plans

CA Practice Partner

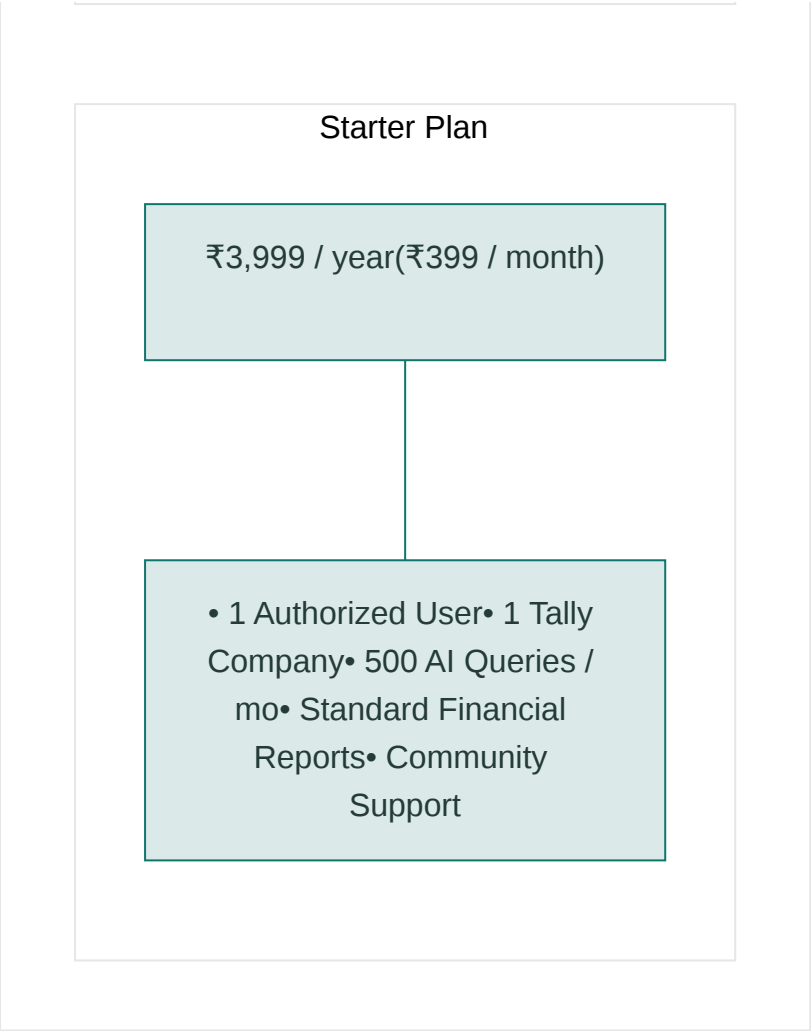
₹17,999 / year(Advisory License)

• Up to 10 Practice Users• Up to 20 Client Companies• Multi-Entity Switcher• Automated Batch MIS• Dedicated Account Manager

Business Growth Plan (Recommended)

₹9,999 / year(Most Popular)

• Up to 5 Team Members• Up to 3 Tally Companies• Unlimited AI Queries• 15-Minute Delta Sync• Priority Email Support



17.0 Competitive Benchmarking & Differentiators

17.1 Comparative Matrix: E-Munshi vs. Riko AI

Capability Dimension	Riko AI	E-Munshi (Pilot v1.0)	E-Munshi Strategic Advantage
AI Conversational Analytics	✅ Included	✅ Included	**Parity** — high-speed natural language interface.
TallyPrime & ERP 9 Support	✅ Included	✅ Included	**Parity** — standard HTTP XML port 9000 extraction.
Mathematical Determinism	Undisclosed	✅ 100% Guaranteed	**E-Munshi Leads:** Formally documented zero-LLM math guarantee prevents hallucinated figures.
Data Freshness Provenance	Partial	✅ Full Transparency	**E-Munshi Leads:** Every card and response displays exact sync time and source report.

Capability Dimension	Riko AI	E-Munshi (Pilot v1.0)	E-Munshi Strategic Advantage
Database Multi-Tenancy	Standard App Auth	✅ PostgreSQL RLS	**E-Munshi Leads:** Defense-in-depth database-enforced row isolation.
Modern Web Architecture	Legacy SPA	✅ Next.js 15 App Router	**E-Munshi Leads:** Unmatched Core Web Vitals, organic SEO acquisition pages, fast initial paint.
Entry Subscription Price	₹5,000 / year	**₹3,999 / year**	**E-Munshi Leads:** 20% more cost-effective for initial enterprise adoption.
Trial Duration	7 Days	**14 Days**	**E-Munshi Leads:** Double the trial window to allow full month-end close evaluation.

18.0 Risk Register & Mitigation Strategy

18.1 Technical, Operational & Adoption Risks

Risk Code	Identified Risk Event	Severity	Probability	Concrete Mitigation Strategy
R-01	Windows Defender / Endpoint Antivirus flags unsigned connector executable.	**HIGH**	**HIGH**	Obtain Microsoft Authenticode Code Signing Certificate; provide an automated diagnostic self-check utility with visual whitelist guides.
R-02	High-volume Tally databases (>100k vouchers) induce timeout during initial sync.	**HIGH**	**MEDIUM**	Implement chunked batch extraction (1,000 vouchers per batch) using TDL envelope pagination; display progress bar in Windows tray.
R-03	Tally hosted in remote cloud VM environments with restricted port access.	**MEDIUM**	**MEDIUM**	Ensure connector executes locally on the virtual desktop hosting Tally; communicate localhost-only requirements in pre-onboarding documentation.
R-04	Malicious prompt injection embedded within Tally ledger or voucher names.	**HIGH**	**LOW**	Enforce strict input sanitization; wrap all master names in CDATA blocks; execute classification on isolated JSON key-value tokens.

Risk Code	Identified Risk Event	Severity	Probability	Concrete Mitigation Strategy
R-05	External LLM API disruption or pricing volatility.	**MEDIUM**	**LOW**	Implement LiteLLM abstraction layer allowing 2-line configuration failover between Google Gemini, Anthropic Claude, and OpenAI endpoints.

18.2 Assumptions & Prerequisites

1. Enterprise host machines execute 64-bit Windows 10 or Windows 11 with continuous outbound internet connectivity.
2. Tally is configured with its standard HTTP server enabled on port 9000 (`Tally.ini` configuration: `Server = Yes`, `Listen = 9000`).
3. Pilot enterprises operate licensed editions of TallyPrime (v2.x through v5.x) or Tally.ERP 9 Release 6+. (Educational editions with locked date ranges are unsupported).

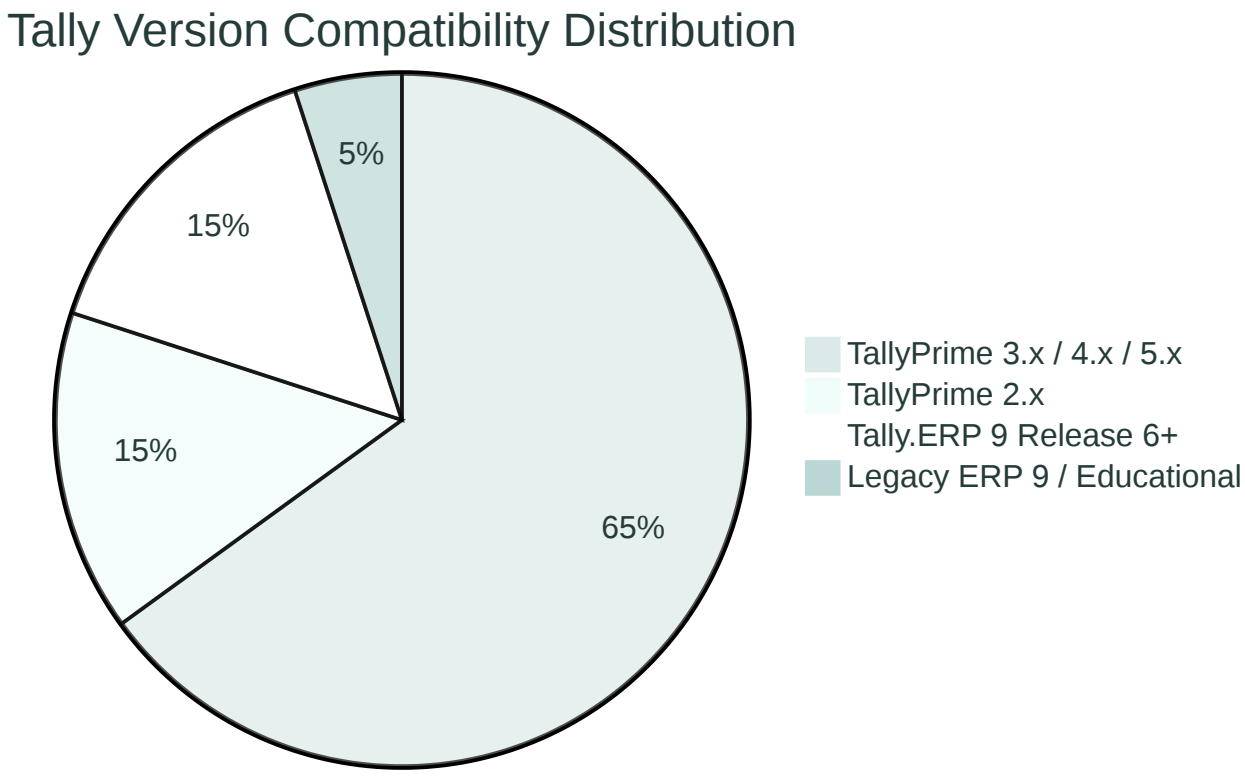
19.0 Pre-Development Governance & Blocker Resolution

> [!CAUTION] **Blocking Governance Items:** The following five decisions must be formally closed prior to commencing Week 1 development to preserve the 30-day delivery window:

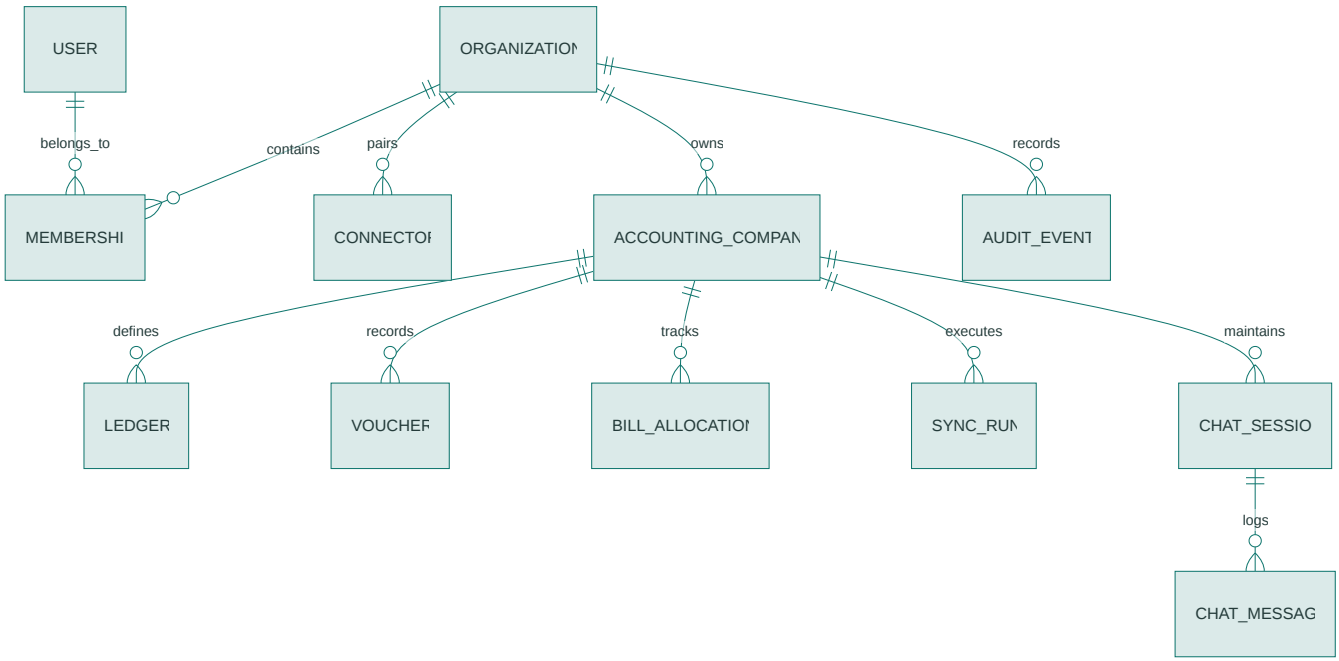
Decision ID	Critical Question	Impact on Sprint Architecture	Owner	Status
Q-01 🔴	**Pilot Tally Versions:** Which exact releases are operated by the 3 pilot clients?	Determines specific TDL envelope compatibility and bill-allocation field tags.	Product Owner	**PENDING**
Q-02 🔴	**Host Environment:** Is Tally installed on local office PCs or remote cloud VMs?	Informs connector packaging, proxy handling, and installation documentation.	Lead Architect	**PENDING**
Q-03 🔴	**Voucher Density:** What is the maximum voucher count across the pilot enterprises?	Sets initial sync batch sizes, memory limits, and timeout boundaries.	Lead Architect	**PENDING**
Q-04 🔴	**Multi-Company Needs:** Do pilot clients require immediate consolidated grouping?	Directs database schema organization and UI company switcher complexity.	Product Owner	**PENDING**
Q-05 🔴	**Reference Standards:** Which native Tally reports represent accepted ground truth?	Defines the acceptance test suite for mathematical reconciliation.	CA Advisor	**PENDING**

20.0 Appendices & Reference Material

Appendix A: Supported Tally Versions



Appendix B: Relational Database Schema (Entity Relationships)



Appendix C: Core API Route Specification

```
Authentication & Identity:
  POST   /api/v1/auth/login                  - Authenticate user credentials
  POST   /api/v1/auth/google                 - Google OAuth PKCE callback
  POST   /api/v1/auth/refresh                - Rotate JWT access token
  DELETE /api/v1/auth/logout                 - Terminate session & clear tokens

Organizations & Tenancy:
  GET    /api/v1/organizations               - List accessible organizations
  POST   /api/v1/organizations               - Provision new organization entity
  GET    /api/v1/companies                   - List connected Tally companies

Connector Lifecycle:
  POST   /api/v1/connectors/pair-init        - Handshake with 6-digit pairing
code
  GET    /api/v1/connectors/status           - Fetch connector heartbeat &
status
  POST   /api/v1/sync/delta                  - Ingest encrypted delta voucher
batch

Financial Statements & Intelligence:
  GET    /api/v1/reports/trial-balance       - Compile deterministic Trial
Balance
  GET    /api/v1/reports/profit-loss         - Compile deterministic P&L
  GET    /api/v1/reports/receivables-aging   - Compile 4-tier aging breakdown
  GET    /api/v1/reports/ledger-statement    - Fetch itemized ledger running
statement

Conversational Intelligence (SSE):
  POST   /api/v1/chat/sessions               - Create conversational session
  POST   /api/v1/chat/sessions/:id/messages  - Submit query & receive streamed
SSE
```

Appendix D: Pre-Release Quality Verification Checklist

- [ ] Automated multi-tenant penetration tests executed; zero cross-tenant record leakage.
- [ ] JWT token rotation and 15-minute expiration validated.
- [ ] All inputs validated via strict Pydantic v2 schemas; SQL injection tests return 0 vulnerabilities.
- [ ] Zero-LLM math guarantee verified: All mathematical operations executed exclusively via Python `Decimal`.
- [ ] `BLOCK_WRITE=1` compile-time flag confirmed on Windows desktop connector.

- [ ] Trial Balance debit and credit totals match Tally native output to 0.00 difference.
- [ ] Profit & Loss Net Profit matches Tally native statement across 3 consecutive closing periods.
- [ ] Receivables aging breakdown matches manual Tally Bills Outstanding report.
- [ ] Standalone connector installer verified on clean Windows 10 and 11 environments.
- [ ] Reconnection resilience tested under simulated broadband network disconnects.
- [ ] High-volume stress test verified with 100,000 vouchers without process failure.
- [ ] Sentry real-time error logging and automated daily database backups verified.

Appendix E: Terminology Glossary

Term	Formal Definition
TallyPrime / ERP 9	The market-standard enterprise accounting software utilized across India by over 7 million organizations.
TDL (Tally Definition Language)	The proprietary XML-based querying language utilized to extract internal collections from Tally.
Port 9000	The native HTTP XML server listening port provided within Tally installations for automated communication.
ALTEREDID	Tally's internal incremental transaction sequence identifier, used as an immutable watermark for delta sync.
Row-Level Security (RLS)	A hardware-grade database security control within PostgreSQL enforcing tenant segregation at the SQL execution layer.
Next.js 15 App Router	Modern full-stack React framework providing hybrid server-rendering (SSR/SSG/ISR) optimized for Core Web Vitals and organic search discoverability.
Deterministic Computation	Algorithmic code execution that guarantees identical, verifiable mathematical outputs without probabilistic variance.
DPDP Act 2023	India's Digital Personal Data Protection Act governing enterprise compliance, user consent, and data sovereignty.

21.0 Document Approval & Sign-Off

The signatures below confirm review and formal authorization of this Product Requirements Document as the governing specification for the E-Munshi 30-Day Pilot Development Sprint.

Organizational Role	Designee Name	Sign-Off Status	Date Certified
Client Sponsor / Product Owner	_____	☐ Approved	_____ / _____ / 2026
Engineering Lead	Mr. Aaquib Nawab	☐ Approved	_____ / _____ / 2026
Chartered Accountant Advisory	_____	☐ Approved	_____ / _____ / 2026

Document Reference: PRD-EMUNSHI-2026-V1.0
Classification: Confidential — Commercial Pilot Deployment
Companion Architectural Blueprint: E-Munshi Product Architecture Document